No-Regret is Not No-Harm: Bandits with Consumable Action Sets

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Abstract

Sublinear regret is the standard justification for deploying sequential decision-making systems. We show that this justification is vacuous when actions consume the action set. We study bandits in which pulling an arm may destroy it, and in which each destroyed arm imposes a permanent negative externality on every subsequent reward. We prove that a policy achieving *exactly zero* regret against the best-available-arm benchmark can incur value loss $m\delta E$, unbounded in both the pool size m and the externality magnitude δE, and can drive total value negative while the optimal policy remains positive. The mechanism is that regret is measured against an optimum the learner has itself degraded. We further establish that the parameter governing optimal play, the expected marginal externality $\kappa_a = p_a e_a$, is (i) exactly the quantity that determines whether greedy is optimal, and (ii) subject to an estimation error floor $2\sigma/\sqrt{\min(T, \sum_j 1/p_j)}$ that is *independent of the horizon*, because each arm supplies exactly one before/after transition and the sample budget is capped by consumption rather than time. The quantity that matters most becomes least estimable exactly as it becomes more important. We give a closed-form correction, characterise an exactly-solvable special case, and release an open-source testbed.

1 Introduction

Consider an orchestrator routing tasks across API-backed tools. Some tools are robust; others trip a rate limit, exhaust a quota, or get a key revoked under heavy use, disappearing for the remainder of the session. Tools are not independent: those sharing upstream infrastructure—a vendor, an IP pool, a quota tier—degrade together when one is burned.

Table 1 shows two routers on identical instances. The first has a flawless record: at every step it pulls the highest-value available tool, so its regret against the best-available benchmark is identically zero. It ends on a fallback tool worth 0.470 per call with two tools remaining. The second reports regret of 0.25 per step—and is sitting at 0.782 per call, 66% higher, with four tools remaining.

The greedy router's zero-regret record is not evidence of good behaviour. It is an artefact of a benchmark that degrades in lockstep with the damage being done: every call greedy makes is optimal *given the ecosystem it has already ruined*.

This paper formalises that failure. Our contributions:

- **Theorem 2** (main result). A zero-regret policy can incur value loss $m\delta E$, unbounded in pool size and externality magnitude, and can finish net-negative while the optimum is positive. No-regret is not no-harm.
- **Theorem 1**. Greedy is optimal if and only if $\kappa_a = p_a e_a$ is constant across arms. Neither factor alone is the invariant.
- **Theorem 3**. The externality cannot be estimated to arbitrary precision: the error floor is horizon-independent, because the data-generating process is the consumption itself.
- **Theorem 5**. In the pure-sequencing regime the optimal order is ascending in e; arm values are irrelevant.
- A closed-form correction (ECI), a rollout policy reaching 99.5–99.9% of optimum, and an open-source testbed with one regression test per theorem.

Table 1: Two routers, identical seed and tools. The policy that looks worse by the standard metric achieves the better outcome.

router	step	realised	regret	tools left
greedy	0	1.150	0.0000	6
greedy	2	0.844	0.0000	4
greedy	7–13	0.470	**0.0000**	**2**
ECI	0	1.150	0.0000	6
ECI	1–4	0.988	0.0500	5
ECI	5–13	**0.782**	0.2500	**4**

2 Related work

Rotting bandits [4, 5, 6] model rewards that decay with the number of pulls (*rested*) or with elapsed time (*restless*); Seznec et al. [6] give a single adaptive-window policy near-optimal for both. Decay there is gradual rather than removal, and critically the learner is assumed to know which regime it is in. **Mortal bandits** [2] have arms with stochastic lifetimes, but expiry is exogenous. **Blocking bandits** make a pulled arm temporarily unavailable. **Sleeping bandits** [3] allow arbitrary exogenous availability. None of these couple arms: destroying one does not change another's reward distribution.

Bandits with knapsacks [1] is the closest structural neighbour: playing an arm consumes resources and the process halts when a budget is exhausted. The budget is global and exogenous, and consumption leaves other arms' rewards untouched. Here consumption permanently degrades *every* future reward. That difference is why the governing quantity is $\kappa = p \cdot e$ rather than a consumption rate.

Bandit learning with positive externalities [7] is the only prior work we are aware of that places externalities in a bandit model. There, satisfied users attract similar users, so externalities are positive and act on the *arrival process*; the authors show UCB incurs linear regret and develop balanced exploration. Our externalities are negative, act on *rewards*, and are triggered by irreversible destruction. The parallel is worth drawing: in both settings a standard algorithm fails badly, indicating that externality structure of either sign breaks classical guarantees.

Work on safe and conservative bandits constrains reward or maintains a baseline while leaving arms available; work on open multi-agent systems models exogenous population churn. Neither addresses action-triggered removal with reward coupling.

3 Setting

Arms $1, \ldots, n$. Arm a has mean reward v_a, destruction probability $p_a \in (0, 1]$, and externality coefficient $e_a \geq 0$. Let S_t be the set of available arms at round t and define the accumulated burden

$$B(S) \;=\; \delta \sum_{i \notin S} e_i.$$

Pulling $a \in S_t$ yields reward $v_a - B(S_t) + \eta_t$ with η_t sub-Gaussian with parameter σ, and destroys a with probability p_a, permanently adding δe_a to the burden. The horizon is T. The value function is

$$V(S,t) = \max_{a \in S} \Big[v_a - B(S) + p_a V(S \setminus \{a\}, t{+}1) + (1 - p_a) V(S, t{+}1) \Big].$$

Define the *expected marginal externality*

$$\boxed{\kappa_a \;=\; p_a \, e_a}$$

the expected permanent burden created by one pull of a.

Benchmark. We measure regret against the best available arm, $\max_{a \in S_t} \big(v_a - B(S_t)\big)$. This is the benchmark deployed systems use, and Theorem 2 is a statement about that practice. Value is measured against the exact optimum $V(S_0, 0)$, computed by dynamic programming over subsets for $n \leq 8$ and by rollout above that.

4 When is greedy optimal?

Theorem 1 (Characterisation)**.** *Greedy—pull* $\arg\max_a v_a$*—is optimal if and only if* κ_a *is constant across arms.*

Proof sketch. Since $B(S)$ and $V(S, t{+}1)$ do not depend on the chosen arm, maximising the Bellman expression is equivalent to maximising $v_a - p_a D_a$ with $D_a = V(S, t{+}1) - V(S \setminus \{a\}, t{+}1)$. Decompose $D_a = \delta e_a L(S,t) + O_a(S,t)$ into a burden and an option component. The burden component contributes $p_a \delta e_a L = \delta \kappa_a L$, which is identical across arms exactly when κ is constant and hence drops out of the argmax.

Sufficiency. Under constant κ, couple the destruction randomness so a pair $\{a, b\}$ receives the same two draws in either order. The surviving-set distribution after both pulls is then identical, and since expected burden per pull is $\delta\kappa$ for both arms, the expected burden paths coincide. The only residual difference is which reward arrives first, giving $v_a - v_b \geq 0$. Iterating sorts any optimal policy into greedy order.

Necessity. The construction of Theorem 2 has $\kappa_H = E > 0 = \kappa_S$ and a strictly positive gap $m\delta E$. □

Two empirical checks. Varying p and e widely while pinning κ constant leaves greedy exactly optimal (0/10 instances broken, gap 0.000%); varying the product breaks it (10/10, gap 5.4–6.3%). Separately, the interchange inequality underlying sufficiency was audited exhaustively over all reachable states: 0 violations in 6,842 ordered pairs under constant κ, versus 56.2% violations when κ varies. The inequality holds precisely when the theorem says greedy is optimal.

The optimality gap grows monotonically with $\mathrm{std}(\kappa)$, from 5.9% to 29.1% across six bins over 240 instances.

5 A zero-regret policy can destroy unbounded value

Theorem 2 (Vacuity of no-regret under consumption)**.** *For any* $M > 0$ *there is an instance on which greedy attains regret exactly* 0 *against the best-available benchmark while* $V^\star - V_{\text{greedy}} \geq M$*. Moreover the loss can exceed* $V^\star$*, making the zero-regret policy net-negative while the optimum is positive.*

Proof. Fix $m \geq 1$, $E > 0$, $0 < \epsilon < \delta E$. Take $n = m{+}1$ arms: one *hot* arm H with $v_H = 1$, $e_H = E$, and m *safe* arms with $v_S = 1 - \epsilon$, $e_S = 0$. Set $p_a = 1$ for all arms and $T = n$, so each arm is pulled exactly once and only the order is chosen.

Greedy pulls H first since $1 > 1 - \epsilon$. H is destroyed, adding permanent burden δE, and each of the m subsequent pulls yields $(1 - \epsilon) - \delta E$:

$$V_{\text{greedy}} = 1 + m(1 - \epsilon - \delta E).$$

Deferring H to last yields $1 - \epsilon$ on each safe pull with no burden, and 1 on the final pull of H, whose burden is charged only to later rounds, of which there are none:

$$V^\star \geq m(1 - \epsilon) + 1.$$

Subtracting, $V^\star - V_{\text{greedy}} = m\,\delta E$, unbounded in m and in δE. Choosing $\delta E > 1$ gives $V_{\text{greedy}} < 0 < V^\star$.

At every round greedy pulls $\arg\max_{a \in S}(v_a - B(S))$, which is the definition of the benchmark; its instantaneous regret is 0 at every state. □

The closed form matches exact dynamic programming to machine precision at all 15 settings tested, $m \in \{1, 2, 4, 8, 12\}$ and $\delta E \in \{0.5, 1, 2\}$. At $m = 12$, $\delta E = 2$: $V^\star = 12.88$, $V_{\text{greedy}} = -11.12$, gap exactly 24.0000, regret 0.0000.

A correction, recorded. An earlier version of this construction claimed the gap grows linearly in T. That is false: with $p = 1$ the pool is exhausted after n pulls, so T beyond n is irrelevant. Under $p < 1$ the gap grows with T only until $T \approx \sum_j 1/p_j$, then saturates—the same sample-budget ceiling that appears independently in Theorem 3. The correct scaling is linear in pool size, which is the meaningful quantity: greedy's error is paid once per subsequent pull.

Interpretation. Greedy maximises v_a, the arm's *private* value. The optimal policy accounts for $\delta\kappa_a(T-t)$, the damage imposed on everything else. Regret is a private-value metric; under externalities private and social value diverge; therefore no-regret guarantees nothing about social outcome.

6 The externality cannot be learned

Theorem 3 (Estimation floor)**.** *For any estimator of δe_i,*

$$\mathrm{SE}(\delta\hat{e}_i) \;\geq\; \frac{2\sigma}{\sqrt{\min\!\big(T,\ \sum_j 1/p_j\big)}},$$

which for $T > \sum_j 1/p_j$ is independent of the horizon.

Proof. All information about e_i enters through the level shift δe_i in the reward process at the moment arm i is destroyed. Conditioning on the destruction time with n_b rounds before and n_a after, the MLE is the difference of pre- and post-destruction sample means, with variance $\sigma^2(1/n_b + 1/n_a)$. By AM–HM, $1/n_b + 1/n_a \geq 4/(n_b + n_a) = 4/N$, with equality iff $n_b = n_a$, giving SE $\geq 2\sigma/\sqrt{N}$. Arm j survives Geometric(p_j) pulls, so $\mathbb{E}[N] = \sum_j 1/p_j$, capped at T. □

All three components verify numerically: the variance formula matches Monte Carlo to within 0.1%; the AM–HM floor is attained exactly at the balanced split (ratio 1.0000); and $\mathbb{E}[N] = \min(T, \sum_j 1/p_j)$ holds including when T binds.

The horizon-independence is stark. At $p = 0.9$, increasing T from 20 to 320 changes RMSE by *zero*—0.2329 at every horizon, because the pool caps the data at 8.8 rounds. At $p = 0.1$ the round count climbs $20 \to 40 \to 71 \to 82.3 \to 83.2$, flattening exactly at the predicted 80.

Corollary 4 (The central tension)**.** *As p rises, $\kappa = pe$ grows, so the externality matters more; simultaneously $N \approx n/p$ shrinks and* SE $\geq 2\sigma\sqrt{p/n}$ *degrades. The parameter governing optimal behaviour becomes less estimable exactly as it becomes more important.*

This is not a sample-complexity result that more data resolves. The data-generating process is the consumption itself: observing arm i's externality requires destroying arm i.

Structure does not rescue it. Suppose $e_a = \langle \psi, z_a \rangle$ for known features $z_a \in \mathbb{R}^k$, $k \ll n$, so each destruction informs a shared k-vector. Per-arm least squares is exactly worthless (identical to greedy to three decimals at every coupling level); feature-based estimation recovers only 17–28% of the available gap; and a *conservative* policy that assumes a uniform upper bound and learns nothing outperforms both at every level. If the mechanism were sample-sharing, capture should rise as k falls. It does not: capture is 4.5% at $k = 1$ and peaks at 30.3% at $k = 5$. At $k = 1$ the externality is nearly uniform, so κ has little dispersion and by Theorem 1 there is nothing to capture. The obstacle is dispersion, not sample size: the externality must vary for the correction to matter, and variation is what makes it hard to estimate.

7 Corrections

7.1 The externality-corrected index

Theorem 1 hands over the correction. Pulling a creates expected permanent burden $\delta\kappa_a$ charged against every remaining round:

$$I(a,t) \;=\; v_a \;-\; \delta\,\kappa_a\,(T-t).$$

Because $B(S)$ is common to all arms it drops out of the ranking, so ECI is a true index: each arm's score depends only on its own parameters and global time, and the ranking is invariant to accumulated damage.

Against exact DP, ECI captures 52%–99% of the optimality gap, with capture improving as coupling strengthens. Its own gap is flat at 1.9–3.1% across the entire dispersion range while greedy's grows five-fold, so ECI's advantage grows without bound in std(κ).

Table 2: The regret and value columns are ordered backwards relative to each other.

policy	value	regret
greedy	14.449	**0.000**
Thompson sampling	15.171	1.341
UCB	13.257	6.136
conservative	17.089	8.160
ECI	**21.214**	10.069

7.2 Rollout

No closed form exists for the option-loss term ECI omits. One step of policy improvement over ECI captures it implicitly, since $V_{\pi_0}(S) - V_{\pi_0}(S \setminus \{a\})$ is exactly the marginal value of holding a. This reaches 99.5–99.9% of exact optimum, and with Monte Carlo policy evaluation runs at $n = 40$ where exact DP is infeasible, adding 3.3% over ECI at strong coupling.

7.3 An exactly solvable case

Theorem 5 (Pure sequencing)**.** *If $p_a = 1$ for all a and $T = n$, the optimal order is ascending in e, and arm values are irrelevant to the optimal order.*

Proof. Every arm is pulled exactly once, so $\sum_a v_a$ is a constant. Total value is $\sum_a v_a - \delta \sum_s e_{a_s}(n - s)$: an arm at position s pays its externality $(n-s)$ times. Minimising requires pairing large e with small $(n-s)$. □

Verified by brute force over all $n!$ orderings, exact at 8/8 instances. Notably ECI is *not* exactly optimal even here—a conjecture we falsified—and the exact rule inverts the usual intuition: when consumption is certain, an arm's value is irrelevant to scheduling; only its externality matters.

8 Experiments

Table 2 compares policies on 20 arms, 60 rounds, 25 seeds.

Thompson sampling is consistently *worse* than greedy under known parameters. Its exploration is not free here: sampling a suboptimal arm can destroy a low-value, high-externality arm. At strong coupling the corrected index nearly triples Thompson's realised value (20.8 vs. 7.2). The mechanisms compose rather than compete—TS handles uncertainty in v, ECI handles consumption—and TS+ECI dominates both.

9 Limitations

The burden model $B(S) = \delta \sum e_i$ is additive and the simplest coupling that exhibits the phenomenon; richer couplings (multiplicative, saturating, networked) are unexplored. Exact optimality results rest on dynamic programming at $n \leq 8$; larger-scale claims rest on rollout. Theorem 2 is a statement about the best-available-arm benchmark specifically—a forward-looking benchmark would not be fooled, but it is also not what deployed systems measure, which is the point. Theorem 1's sufficiency proof relies on a coupling argument whose formalisation under horizon truncation we verify exhaustively rather than prove in full generality.

10 Conclusion

When actions consume the action set, regret is measured against an optimum the learner has already degraded. A policy can hold regret at exactly zero while driving value arbitrarily negative, and the parameter that would let it do better becomes unlearnable precisely as it becomes important, because acquiring each observation destroys the resource being priced. The practical consequence is direct: for irreversible actions, externality costs must be *specified*, not discovered. A crude uniform upper bound outperforms every learned estimator we tested.

Code. Library, all 24 experiments including the falsified ones, and one regression test per theorem: `https://github.com/<user>/consumable-bandits`

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